Video-rate terahertz single-pixel imaging via physics-enhanced deep learning and VCSEL array modulation

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ABSTRACT

Terahertz (THz) single-pixel imaging has emerged as a significant research direction due to its cost-effectiveness, low system complexity, and high scalability. However, practical implementation is hindered by a critical trade-off between imaging speed and quality, stemming from the combined limitations of modulator response speeds and reconstruction algorithms. Addressing this challenge, we propose a physics-enhanced deep learning framework for video-rate THz single-pixel imaging. By utilizing a high-speed vertical-cavity surface-emitting laser array to pump a high-resistivity silicon wafer, we achieve high-speed THz modulation and remove the refresh-rate limitation associated with commonly used DMD-based optical pattern generation. To bridge the gap between hardware addressing constraints and the requirements for complex encoding masks, a low-rank matrix synthesis strategy is integrated into the neural network training to jointly optimize mask distribution and image reconstruction. Both simulation and experimental results demonstrate that this synergistic mechanism outperforms pure data-driven algorithms and traditional basis scanning methods, enabling real-time THz video capture at 50 f/s with a resolution of 32 × 32 pixels. Even at a 25% sampling ratio, the system achieves high-fidelity reconstruction, effectively balancing speed and quality. This hardware–software co-designed approach paves the way for real-time THz applications in industrial and scientific research while establishing a new paradigm for high-speed dynamic imaging across other spectral regimes.

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I. INTRODUCTION

Terahertz (THz) radiation, located within the 0.1–10 THz frequency band, bridges the gap between electronics and photonics. Owing to its unique attributes—including non-ionizing nature, high penetrability through non-metallic materials, and high sensitivity to polar molecules—THz technology has demonstrated immense potential in diverse fields such as non-destructive testing, biomedical imaging,[1] security inspection,[2,3] and material analysis.[4,5] However, the prohibitive cost of high-performance THz transmitters and detectors, and the inherent bottlenecks of focal plane arrays regarding noise suppression, pixel scalability, and fabrication complexity, have severely constrained the efficiency and widespread deployment of conventional imaging systems.[6] In this context, single-pixel imaging (SPI) has emerged as a compelling alternative, offering distinct advantages such as low hardware costs and high system flexibility.[7] Nevertheless, a deeper analysis reveals that current THz SPI technology remains hindered by a fundamental trade-off between imaging speed and reconstruction quality.[8] This efficiency bottleneck stems from cumulative limitations at both the encoding and reconstruction ends. On the encoding side, factors such as restricted spatial modulation rates, the requirement for a

large number of effective patterns,[9,10] and insufficient modulation depth[11] impede efficient information acquisition. On the reconstruction side, conventional algorithms struggle to simultaneously balance computational speed and imaging fidelity under conditions of undersampling, strong noise, or low signal-to-noise ratio (SNR). This dual constraint severely limits overall system performance.

At the encoding stage, which serves as the front-end for information acquisition, the efficiency of spatial pattern modulation is crucial. It directly influences the temporal resolution of SPI. Operating on the paradigm of encode–sample–decode–reconstruct, the total imaging time is inherently governed by the product of the number of measurements (M) and the switching time of a single pattern (t_0). Early implementations employed mechanical patterns, such as metal plates[12] or spinning disks,[13] to modulate THz beams; however, their slow switching speeds proved inadequate for rapid imaging applications. Subsequently, optically pumped semiconductor modulation emerged as the dominant solution.[14,15] Utilizing photo-generated carrier modulators based on materials like Si, SiO_2/Si,[16–18] and conjugated polymer–silicon hybrid structures,[19] these approaches offer modulation rates ranging from kHz to tens of kHz while effectively eliminating mechanical vibrations and alignment complexities. To curtail the number of measurements, researchers have leveraged compressed sensing (CS) with random patterns[20,21] or deterministic orthogonal bases such as Hadamard[22–24] and Fourier patterns,[25] successfully reducing the sampling ratio to the range of 10%–40%. Here, the sampling ratio is defined as $\beta = M/(N \times N)$, representing the ratio of the number of sampled patterns M to the total number of image pixels $N \times N$. For instance, She *et al.* utilized a Digital Micromirror Device (DMD) to implement Fourier encoding for THz waves, achieving high-quality reconstruction at a 10% sampling ratio,[25] while Stantchev *et al.* employed a DMD with Hadamard encoding to demonstrate THz single-pixel video at 6 f/s (fps).[26] However, these schemes remain constrained by the intrinsic refresh rate limit of DMDs (~22 kHz), making it challenging to surpass the frame rates required for smooth video-level imaging.[27] Consequently, the dual factors of sampling quantity and modulation speed persist as the core bottlenecks impeding the efficiency of THz imaging systems.

However, merely increasing the hardware modulation rate is insufficient to fully resolve the challenges of real-time imaging. The reconstruction process itself must keep pace. Particularly under conditions of undersampling or severe noise, the performance of image reconstruction algorithms has emerged as a new efficiency bottleneck. THz SPI typically relies on compressed sensing or basis scanning methods for image recovery.[28] Yet, in scenarios characterized by extremely low sampling ratios or elevated noise levels, iterative CS algorithms often suffer from slow convergence and instability,[29,30] whereas the reconstruction quality of basis scanning methods is highly susceptible to noise and pattern correlation artifacts. In recent years, advances in deep learning have catalyzed new breakthroughs in SPI.[31–33] End-to-end networks, utilizing raw measurements or low-quality initial reconstructions as inputs, have demonstrated the ability to achieve high-quality imaging even at ultralow sampling ratios.[34–36] Furthermore, hybrid schemes combining Hadamard decoding with deep networks have reduced sampling times to ~1/20th of traditional methods and compressed pattern usage to roughly 10% of the total pixel count, while maintaining a Peak Signal-to-Noise Ratio (PSNR) exceeding 20 dB and a Structural Similarity Index (SSIM) above 0.85.[37] Moreover, data-driven networks (e.g., DnCNN,[38] DCAN,[39] and RDN[40]) typically treat image reconstruction as a post-processing task focused on denoising or artifact removal. While DnCNN abandons traditional direct latent image estimation, it uses residual learning and batch normalization to predict noise distribution, enabling efficient signal separation by subtracting background noise from observations. DCAN, in turn, introduces a domain-conditioned channel attention mechanism that adaptively recalibrates convolutional channels via a dual-route feature calibration network, enhancing edge fidelity and structural consistency while suppressing artifacts. However, these approaches remain fundamentally data-driven, relying on neural networks to fit image statistics primarily for enhancement purposes. Currently, deep learning methodologies are evolving from purely data-driven paradigms toward physics-informed frameworks. This evolution involves the joint optimization of pattern generation and reconstruction networks within a system-level architecture[41] and the incorporation of imaging physical priors—such as diffraction,[42,43] modulation transfer functions, and noise models—to enhance reconstruction stability. Despite these advancements, research on deep learning for real-time frame-rate reconstruction of dynamic THz targets remains relatively scarce. In particular, systematic solutions for addressing inter-frame motion blur and maintaining structural coherence under extremely short integration times are still lacking.

In contrast to THz imaging, which remains constrained by modulation hardware speeds, visible-light single-pixel imaging has witnessed the emergence of ultra-high-speed modulation schemes capable of achieving frame rates from thousands to tens of thousands of frames per second (fps). For instance, leveraging the sub-microsecond switching characteristics of LED arrays, Sun *et al.* demonstrated high-speed SPI with 32×32 pixels at frame rates ranging from 1000 to 25 000 fps.[44,45] Recently, we reported a vertical-cavity surface-emitting laser (VCSEL)-based THz single-pixel imaging system that enables high-speed pattern projection under a row-column addressing scheme. While this approach effectively alleviates the refresh-rate limitation of conventional DMD-based systems, the illumination patterns are predefined, and the reconstruction relies on conventional algorithms, which limits the achievable performance under reduced-sampling conditions and noisy environments.[46] Addressing this critical challenge, we propose a THz SPI framework that integrates VCSEL array modulation with physics-enhanced deep learning, termed the VCSEL-driven Constrained Physics Encoding Network (VCPEN), enabling joint optimization of encoding and reconstruction under hardware constraints. By establishing a synergistic closed loop integrating VCSEL photonic modulation with deep learning computation, we successfully demonstrate real-time THz single-pixel video imaging at 50 fps. This framework features significant innovations across three dimensions: On the hardware front, we introduce a VCSEL array as an ultrafast pump source to excite photocarrier dynamics within high-resistivity silicon. This approach removes the refresh-rate limitation associated with commonly used DMD-based optical pattern generation, establishing a robust physical foundation for high-speed encoding. On the computational backend, moving beyond purely

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data-driven paradigms, we design a physics-enhanced network that is strictly matched to the frontend optoelectronic modulation characteristics. By incorporating hardware encoding constraints directly into the network training, we achieve joint optimization of pattern generation and image reconstruction, significantly enhancing reconstruction quality under conditions of low sampling rates and high noise. At the system level, this holistic hardware–software co-design strategy enables the system to maintain superior imaging SNR and structural fidelity even at 50 fps. This methodology effectively bridges the long-standing dual bottleneck between modulation rate and algorithmic performance in THz SPI, offering a systematic physics-enhanced paradigm for ultra-high-speed single-pixel imaging across various spectral bands.

II. SYSTEM SETUP

To address the dual challenge of low encoding speed and poor reconstruction quality in THz SPI, we developed the VCPEN framework. This framework realizes real-time video-rate THz imaging through synergistic innovations across hardware, algorithmic, and system levels.

A. Ultrafast optoelectronic modulation hardware based on VCSEL array

The schematic of the proposed VCPEN THz SPI system is illustrated in Fig. 1. The core spatial light modulation unit is a 32 × 32 VCSEL array operating at a center wavelength of 850 nm, with an effective emission area of 62 × 62 mm^2. The row-column addressing of the array is driven by high-speed signals generated from an FPGA control board (DFPGL50G), which connects to the driver circuit via 64 I/O interfaces to achieve rapid loading and refreshing of the illumination patterns. In the THz optical path, a modular source (TeraSense) with a center frequency of 100 GHz and an output power of ~80 mW is employed. The emitted THz beam is collimated by an off-axis parabolic mirror (OAP) and reflected by a mirror coated with an Indium Tin Oxide (ITO) thin film (sheet resistance ~15 Ω/sq) onto the surface of a high-resistivity silicon modulator. The modulator utilizes a high-resistivity silicon wafer with a thickness of 500 μm, a resistivity exceeding 10 000 Ω cm, and a carrier lifetime of 12 μs. A laser-engraved stainless steel stencil, serving as the imaging target, is placed in close contact with the backside of the silicon wafer. To achieve efficient spatial modulation, the encoding patterns projected by the VCSEL array are focused onto the silicon surface by a Fresnel lens (f = 120 mm, diameter 250 mm, thickness 2 mm), covering an illumination area of ~25 × 25 mm^2. The incident pump light induces a spatial distribution of photocarriers within the silicon, thereby dynamically modulating the transmission coefficient of the THz wave via free-carrier absorption. The modulated THz signals are subsequently focused by a 2-in. TPX lens (f = 100 mm) into a single-pixel detector. The detector employs an InP-based Schottky barrier diode with a bandwidth of 70–110 GHz and a sensitivity of 1000 V/W. The output voltage signals are synchronously recorded by a data acquisition system (NI USB-6361) at a sampling rate of 2 MS/s. To quantify the physical modulation performance of the system, static characterization tests were conducted. Under full illumination with a total optical power of 1.25 W, the detector output voltage shifted from −0.75 V (Dark/OFF state) to −0.525 V (Bright/ON state), corresponding to a THz modulation depth of ~30%.

B. VCPEN data flow framework and acquisition strategy

To achieve efficient encoding and reconstruction, we constructed a physics-enhanced data flow processing framework. This framework deeply integrates the physical imaging model into the

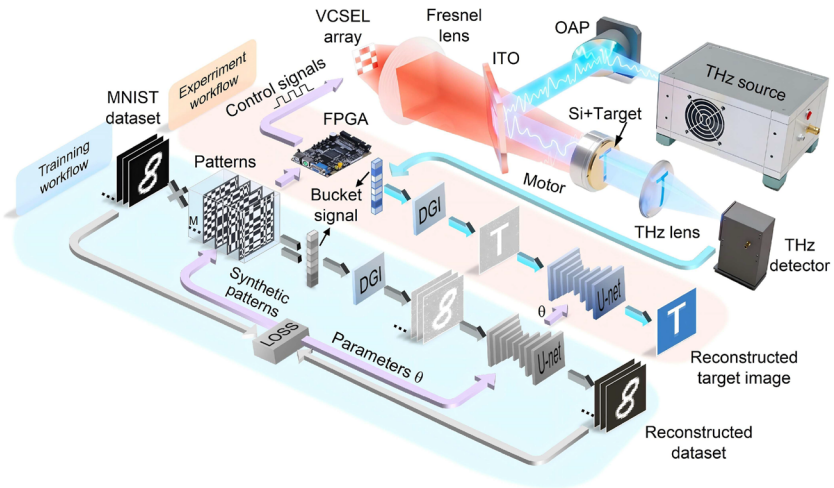


FIG. 1. Schematic of the VCPEN THz SPI framework featuring hardware–software synergy. In the hardware implementation, a 100 GHz THz beam is collimated by an OAP and directed via an ITO mirror onto a high-resistivity silicon modulator. An FPGA-driven 32 × 32 VCSEL array projects high-speed pump patterns through a Fresnel lens, dynamically modulating THz transmission via free-carrier absorption before collection by a single-pixel detector. The framework's operational logic is divided into two workflows. In the training workflow, the pattern synthesis strategy and U-Net parameters θ undergo joint end-to-end optimization by minimizing the loss between reconstructed MNIST images and the ground truth. In the experimental workflow, the optimized patterns are deployed for hardware signal acquisition, followed by target image reconstruction using the pre-trained U-Net.

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computational pipeline, realizing a complete link from encoding to reconstruction through three synergistic modules: First, to accommodate the specific control mechanism of the VCSEL array, we employ a pattern generation module based on Rank-1 Matrix synthesis to define a physically compatible sampling strategy on a low-rank manifold, guiding the hardware to execute differential acquisition; subsequently, the acquired signals are processed using fast Differential Ghost Imaging (DGI) to obtain a preliminary image; finally, a U-Net leveraging data-driven priors performs a nonlinear mapping to recover high-frequency details and eliminate residual noise.

1. Physics-constrained rank-2 pattern synthesis and decomposition

The VCSEL array adopted in this study employs a row-column matrix addressing scheme similar to conventional LED matrices, using $H + W$ control signals for $H \times W$ pixels. As shown in Fig. 2(a), an example of a 2×2 pixel LD array and the circuit of the designed driver are presented. Although it cannot address individual pixels independently like DMD devices, it reduces circuit complexity and power consumption that would result from using $H \times W$ independent drivers. Consequently, the projected optical pattern is formed by the outer product of a row and a column vector, resulting in a Rank-1 matrix. In linear algebra, a matrix has a rank of 1 if all its rows are linearly dependent on a single basis vector, which, in our case, constrains the modulation patterns to a low-dimensional subspace. To maintain sufficient coding complexity while satisfying this hardware constraint, we propose a Rank-2 pattern matrix synthesis strategy: each logical pattern is formed by optimizing two 1D vectors and combining them into a Rank-2 matrix that can be decomposed into two Rank-1 components for projection.

To make the pattern synthesis process more intuitive, Fig. 2(b) illustrates the complete workflow using a 5×5 example. First, two 1D binary vectors are generated to represent the row and column activation states. These vectors are then used to construct two rank-1 sub-patterns through outer-product operations. By combining the two sub-patterns, a logical Rank-2 pattern is obtained, which provides a richer spatial structure than a single rank-1 pattern. However, because the VCSEL array is driven by row-column addressing, this logical Rank-2 pattern is not directly projected in a single step. Instead, it is decomposed into sequential hardware-compatible projections. Furthermore, complementary differential acquisition is introduced to suppress source fluctuation and measurement noise. As a result, each logical Rank-2 pattern is implemented experimentally through four sequential projection steps. This procedure establishes the connection between the learnable logical mask, the hardware-compatible projection sequence, and the final differential measurement used for reconstruction. In the synthesis part, the mth logical pattern $P_m^{(\mathrm{DL})}$ is synthesized from two learnable 1D vectors (row vector $\mathbf{r} \in \mathbb{R}^{1\times H}$ and column vector $\mathbf{c} \in \mathbb{R}^{1\times W}$). These two vectors generate two Rank-1 matrices via specific outer-product operations, which are then summed to yield the Rank-2 matrix $P_m^{(\mathrm{DL})}$, mathematically expressed as

$$P_m^{(\mathrm{DL})} = \underbrace{\mathbf{r}_b[1-\mathbf{c}_b]}_{\text{Rank}-1\ \text{Matrix A}} + \underbrace{[1-\mathbf{r}_b]\mathbf{c}_b}_{\text{Rank}-1\ \text{Matrix B}}, \tag{1}$$

where $\mathbf{r}_b$ and $\mathbf{c}_b$ denote the binarized vectors (the details of the binarization method are provided in the supplementary material). This strategy not only significantly reduces the optimization parameter complexity from $\mathcal{O}(H \times W)$ to $\mathcal{O}(H + W)$ but also ensures the physical compatibility between the algorithm pattern and the hardware architecture while maintaining sufficient coding complexity, providing a key foundation for subsequent decomposition and projection.

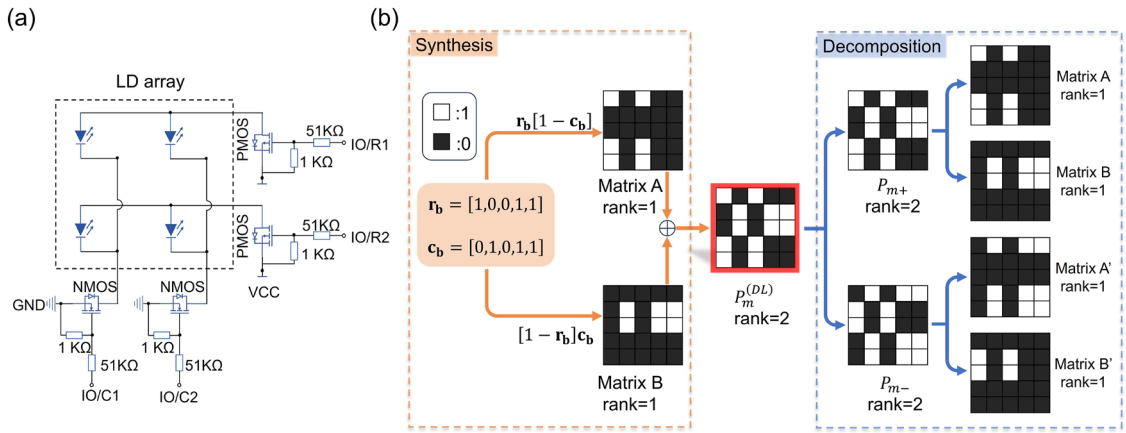


FIG. 2. (a) Schematic of the row–column addressing circuit of the LD (VCSEL) array. (b) Schematic of the physics-constrained Rank-2 pattern synthesis and decomposition for the VCPEN system. For illustration, a 5×5 pattern is employed. The flow chart illustrates the derivation of Rank-1 fundamental components from 1D vectors for Rank-2 pattern construction and their subsequent translation back into hardware-compatible Rank-1 projection sequences. The synthesis process demonstrates how two complementary Rank-1 matrices (Matrix A and Matrix B) are obtained via the operation of two 1D binary vectors ($\mathbf{r}_b$ and $\mathbf{c}_b$) and then summed to generate the Rank-2 logical pattern $P_m^{(\mathrm{DL})}$. The decomposition process depicts the hardware implementation strategy: the generated pattern $P_{m+} = P_m^{(\mathrm{DL})}$ and its inverse phase $P_{m-} = 1 - P_m^{(\mathrm{DL})}$ are each decomposed into two fundamental Rank-1 matrices (A/B and A′/B′, respectively).

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In the actual pattern projection and data acquisition stage, the system implements a decomposition projection strategy. To suppress environmental light and device common-mode noise, we adopt a differential measurement strategy, wherein for each generated pattern $P_m^{(\mathrm{DL})}$, its positive phase P_{m+} and negative phase P_{m-} ($P_{m+} = P_m^{(\mathrm{DL})}, P_{m-} = 1 - P_m^{(\mathrm{DL})}$) are projected sequentially. Subsequently, based on the inverse process of the aforementioned mathematical definition [Eq. (1)], each complementary pattern is decomposed into two physically independent Rank-1 matrices (Matrix A/B for P_{m+} and Matrix A′/B′ for P_{m-}). These are then converted into row and column timing control signals for the FPGA (timing diagrams are detailed in the supplementary material) to drive the VCSEL array sequentially. Consequently, the complete projection of a single logical pattern $P_m^{(\mathrm{DL})}$ requires a four-step projection strategy executed in four sequential temporal steps: two Rank-1 projections (A/B) for P_{m+} and two Rank-1 projections (A′/B′) for P_{m-}. This design, combining low-rank synthesis with complementary differential measurement, achieves high-dimensional information acquisition at the physical level with minimal temporal cost, while effectively suppressing common-mode noise and power fluctuations. Furthermore, the system allows for a flexible dynamic balance between imaging frame rate (up to 50 fps) and signal-to-noise ratio by adjusting the dwell time of individual patterns. Finally, the single-pixel detector integrates the signals to obtain the corresponding 1D bucket signal sequence,

$$D_m = \sum_{x=1}^{H} \sum_{y=1}^{W} P_m^{(\mathrm{DL})}(x,y)\, O(x,y). \tag{2}$$

2. Physics-informed DGI reconstruction

Upon acquiring the bucket signal sequence $\{D_m\}$ of length M, we utilize the Differential Ghost Imaging (DGI) algorithm to reconstruct this 1D temporal signal sequence into a 2D spatial distribution. As an efficient non-iterative method, DGI rapidly produces an initial reconstruction by evaluating the statistical correlation between the modulation patterns and the corresponding detected intensity signals. Its mathematical formulation is expressed as

$$O_{DGI} = \left\langle P_m^{(\mathrm{DL})} D_m \right\rangle - \frac{\left\langle P_m^{(\mathrm{DL})} \right\rangle}{\langle S_m \rangle} \langle S_m D_m \rangle, \tag{3}$$

where $\langle\cdot\rangle$ denotes the ensemble average, mathematically approximated as $\left\langle P_m^{(\mathrm{DL})} \right\rangle = \frac{1}{M}\sum_{m=1}^{M} P_m^{(\mathrm{DL})}$. Here, $\langle S_m \rangle = \frac{1}{M}\sum_{m=1}^{M} S_m$ represents the mean total intensity across all patterns, while $S_m = \sum_{x,y} P_m^{(\mathrm{DL})}(x,y)$ denotes the total optical intensity (or the number of transmissive pixels) of the mth encoding pattern. Through these averaging and normalization operations, DGI effectively suppresses system common-mode noise. Its reconstruction result serves as an effective physical prior, providing crucial initial guidance for the subsequent neural network's optimization and fine reconstruction.

3. Nonlinear detail enhancement via U-Net

Following the initial reconstruction by DGI (O_{DGI}), we provide this result as input to a classic four-level U-Net to further enhance nonlinear details (see the supplementary material for the detailed U-Net architecture). Leveraging deep convolutional features, the network captures high-frequency details and effectively suppresses residual speckle noise and aliasing artifacts inherent to DGI, thereby reconstructing a high-fidelity final image O_{DL},

$$O_{DL} = f_{\mathrm{U-Net}}(O_{DGI}; \theta), \tag{4}$$

where θ denotes the set of learnable parameters within the U-Net.

To achieve globally optimal coordination between encoding and decoding, we incorporate both the pattern generation parameters and the image reconstruction parameters into a unified end-to-end co-optimization framework. During the training phase, utilizing the dataset $\mathscr{D}_{train} = \{x^k | k = 1, 2, \ldots, K\}$, we minimize the L_2 norm loss between the reconstructed image O_{DL} and the ground truth O via the backpropagation algorithm. This process jointly solves for the optimal network parameters θ^* and the optimal set of physical patterns $\mathscr{P}_{\mathrm{DL}}^*$,

$$\{\theta^*, \mathscr{P}_{\mathrm{DL}}^*\} = \underset{\theta \in \Theta,\ \mathscr{P}_{\mathrm{DL}} \in \mathscr{P}}{\arg\min} \left\| O_{DL} - O \right\|^2. \tag{5}$$

This hybrid architecture, combining physical model constraints with a deep implicit prior, not only inherits the physical stability of ghost imaging but also fully leverages the powerful nonlinear expressive capability of data-driven approaches in resolving ill-posed inverse problems.

III. RESULTS AND DISCUSSION

This section validates the proposed network framework and constrained model and provides a comparative performance analysis. We first conduct quantitative evaluations through simulation experiments against other reconstruction algorithms. Subsequently, experimental imaging results from the actual system are presented to verify the method's effectiveness.

A. Simulation

In the simulation phase, the publicly available MNIST handwritten digits dataset was used for both training and evaluation. Specifically, 60 000 images were randomly selected for the training set, and another 10 000 images were used for the test set. All images were normalized to grayscale and uniformly resized to 32 × 32 pixels. The initial training parameters were set as a learning rate of 0.001 and a batch size of 512. The model training was executed on a computing platform equipped with an Intel Core i9-10900K CPU, 64 GB RAM, and an NVIDIA GeForce RTX 3090 GPU. To quantitatively evaluate the reconstruction performance, we selected the PSNR and SSIM as evaluation metrics, assessing the imaging quality from the two dimensions of pixel error and structural fidelity, respectively.

We evaluated our VCPEN model's performance in two key dimensions: noise robustness and generalization under low sampling rates. For fair performance benchmarking, all comparison algorithms employed Walsh–Hadamard patterns for sampling and adopted the identical differential projection acquisition strategy and were combined with three representative reconstruction schemes: the traditional Hadamard reconstruction method based on physical inversion (HSI) and two purely data-driven neural networks, DnCNN and DCAN, which are dedicated to post-processing and enhancing the initial DGI reconstruction results.

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To investigate the system's noise robustness, we selected the 32×32 Chinese character as a test target. This target, which is not included in the MNIST dataset, was specifically chosen for its more complex stroke structure compared to the MNIST digits. In this analysis, additive white Gaussian noise is applied directly to the ground-truth images. The SNR is defined based on the signal power of the target image as $SNR_{dB} = 10\lg(P_{image}/P_{noise})$, where P_{image} and P_{noise} represent the power of the clean image and the added noise, respectively. We fixed the number of patterns at $M = 512$ ($\beta = 50\%$) and tested six noise scenarios ranging from 0 to 20 dB, plus a noise-free condition. The selection of 512 patterns is physically justified: based on the modulator's carrier lifetime of 12 μs, a dwell time of 20 μs is required per projection step to ensure steady-state modulation. Combined with the four-step projection strategy, sampling with 512 patterns precisely corresponds to a video frame rate of ~25 fps. Figure 3 illustrates the performance of the four reconstruction methods under these conditions across different noise levels. The VCPEN framework demonstrates superior visual quality and significantly stronger noise suppression at all noise levels. Notably, even in the extreme 0 dB strong noise environment, VCPEN manages to recover a recognizable character structure. In contrast, images reconstructed by the traditional HSI algorithm exhibit pronounced residual noise due to its inherent lack of a denoising mechanism. While DnCNN and DCAN networks partially improve denoising performance, they suffer from excessive smoothing and insufficient sharpness, leading to the loss or blurring of critical stroke details of the character. Consequently, their performance in terms of structural fidelity and overall reconstruction accuracy remains significantly lower than that of the VCPEN framework.

In further quantitative experiments, 200 image samples were randomly selected from the test set. The average PSNR and SSIM of each method were calculated under six noise levels ranging from noise-free to 20 dB to evaluate the stability and robustness of the model under different noise conditions, with results shown in Fig. 3. As the noise intensity increases, the reconstruction quality of all methods shows a decreasing trend. Among them, the performance curve of the traditional HSI algorithm decays most dramatically, indicating its high sensitivity to noise. It is noteworthy that although data-driven methods such as DnCNN and DCAN improved noise resistance compared to HSI, the VCPEN model consistently maintained significant performance leadership across all noise levels. Even in the extreme 0 dB noise environment, the model's average PSNR still exceeded 20 dB, and SSIM remained above 0.8; under ideal noise-free conditions, the average PSNR reached 25.12 dB, and SSIM reached 0.8923. These quantitative results confirm that, in low-SNR regimes, incorporating physical constraints effectively enhances feature extraction, offering a crucial complement to the performance of purely data-driven networks.

In a separate simulation experiment, we validated the model's low-sampling reconstruction and cross-domain generalization using multiple classes of noise-free 32×32 images—including letters, symbols, Chinese characters, and geometric figures—as out-of-training test samples. Experiments were conducted using two configurations: 512 patterns ($\beta = 50\%$) and 256 patterns ($\beta = 25\%$). The 256 pattern configuration represents the minimum count necessary for 50 fps high-speed video, given the fixed 20 μs dwell time per projection step. The results are shown in Fig. 4. The traditional HSI algorithm produced obvious mosaic artifacts and residual noise under undersampling conditions. Although purely data-driven networks like DnCNN and DCAN effectively removed noise, they struggled to recover fine details, resulting in blurred symbol edges and the loss of subtle structures. In contrast, the VCPEN model, benefiting from the synergistic drive of physics and data, obtained clear, structurally complete reconstructed images across both sampling configurations. Table I quantitatively corroborates this advantage: our method achieves the highest SSIM across all categories and sampling rates. Crucially, under the challenging low-sampling condition

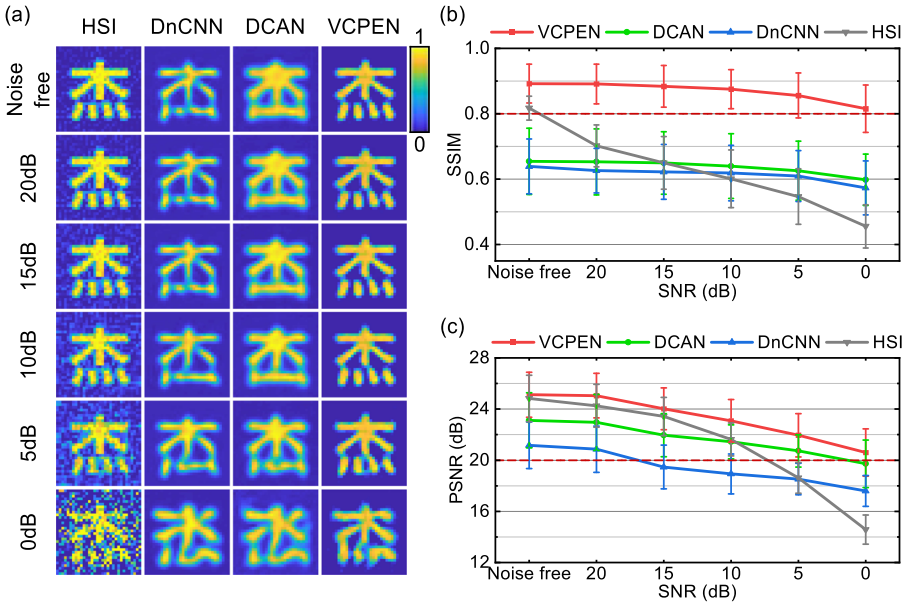


FIG. 3. Noise robustness under varying signal-to-noise ratios. (a) Visual comparison of 32×32 test target (Chinese character) reconstructions by HSI, DnCNN, DCAN, and VCPEN from noise-free conditions to 0 dB SNR. The sampling number is fixed at 512 ($\beta = 50\%$), and the color bar denotes normalized intensity. (b) Average SSIM and (c) PSNR over 200 test images vs SNR, with error bars indicating standard deviation. The red dashed lines mark the high-fidelity thresholds (SSIM = 0.8, PSNR = 20 dB).

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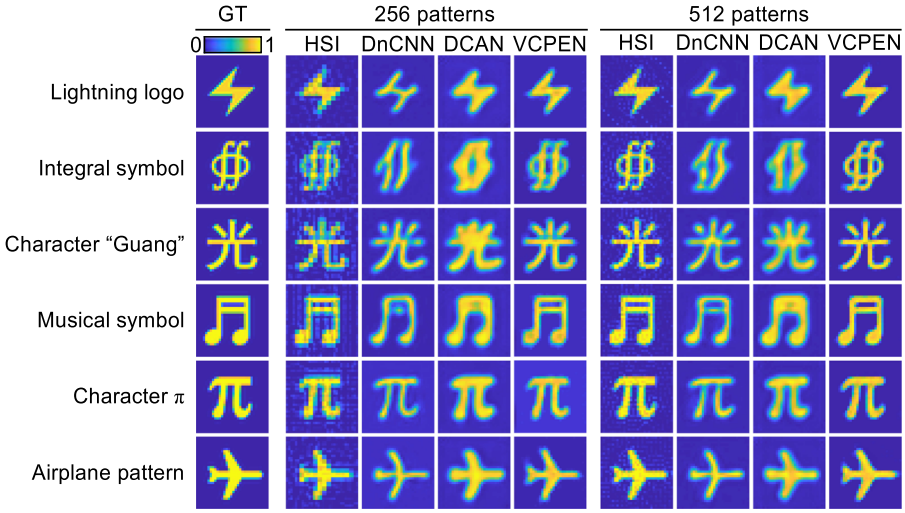


FIG. 4. Simulation reconstruction results of various targets under different sampling ratios. The test samples consist of 32×32 pixel symbols, characters, and geometric patterns that were not involved in the training stage. Columns from left to right: Ground Truth (GT), followed by reconstruction results from HSI, DnCNN, DCAN, and VCPEN under 256 patterns ($\beta = 25\%$) and 512 patterns ($\beta = 50\%$). The color bar indicates the normalized intensity.

TABLE I. Reconstruction performance comparison under 256 and 512 illumination patterns. Boldface denotes the best performance in each metric.

Number of patterns		256 patterns ($\beta = 25\%$)				512 patterns ($\beta = 50\%$)			
Target\methods		HSI	DnCNN	DCAN	VCPEN	HSI	DnCNN	DCAN	VCPEN
Lightning logo	PSNR (dB)	21.03	18.51	18.86	**23.94**	25.35	19.84	18.56	**26.75**
	SSIM	0.57	0.46	0.32	**0.63**	0.63	0.54	0.30	**0.91**
Integral symbol	PSNR (dB)	16.97	13.96	12.08	**17.02**	**22.44**	14.20	13.97	19.80
	SSIM	0.45	0.35	0.28	**0.77**	0.61	0.41	0.37	**0.87**
Character "Guang"	PSNR (dB)	16.90	16.17	12.08	**18.63**	22.50	16.77	14.64	**22.99**
	SSIM	0.54	0.58	0.36	**0.86**	0.71	0.59	0.46	**0.91**
Musical symbol	PSNR (dB)	18.76	15.07	15.36	**20.34**	**25.70**	16.32	15.36	23.29
	SSIM	0.57	0.64	0.46	**0.59**	0.77	0.54	0.52	**0.92**
Character π	PSNR (dB)	18.35	16.10	16.95	**18.85**	**24.77**	16.35	18.01	22.52
	SSIM	0.49	0.41	0.44	**0.69**	0.68	0.51	0.50	**0.91**
Airplane pattern	PSNR (dB)	19.26	16.47	18.11	**21.07**	**25.29**	17.11	18.28	24.32
	SSIM	0.52	0.39	0.42	**0.76**	0.65	0.54	0.44	**0.91**

of $\beta = 25\%$, VCPEN simultaneously achieved the highest PSNR and SSIM metrics, confirming the system's exceptional robustness in high-speed, low-sampling, and cross-type imaging tasks.

In summary, the simulation experiments conclusively validated the robustness and cross-domain generalization capability of the VCPEN framework, even under conditions of low sampling rates and severe noise. This comprehensively confirms the theoretical effectiveness of the proposed method in complex imaging scenarios, thus laying a solid foundation for its subsequent hardware deployment and experimental verification within the practical THz system.

B. Experiment

To validate the actual imaging performance of the proposed method, the experimental section utilized the same network structure and comparison algorithms (HSI, DCAN, and DnCNN) as the simulation phase. Multiple metal stencil samples with different structural features were used as targets. Figure 5 displays the 32×32 pixel reconstruction results of each method under 256 patterns ($\beta = 25\%$) and 512 patterns ($\beta = 50\%$) conditions. The system executes the four-step projection strategy, with a single projection time set to 20 μs. Under these parameters, the single-frame imaging time for 512 patterns is only 0.04 s (corresponding to 25 fps), while 256 patterns further compress this time to 0.02 s (corresponding to 50 fps). This temporal achievement validates the effective combination of pattern constraints and matrix addressing control in enhancing temporal efficiency, confirming the system's potential for real-time video-rate imaging.

A photograph of the experimental setup is shown in Fig. 5, where the main components and optical path are labeled. From the qualitative results in Fig. 5, the reconstruction quality of the HSI method is limited by environmental noise, showing significant noise residue in the background. Purely data-driven networks like DCAN and DnCNN, while significantly suppressing random noise, tend to over-smooth the images when processing real optical path data, as observed in simulations, leading to blurring of the sharp edges of the

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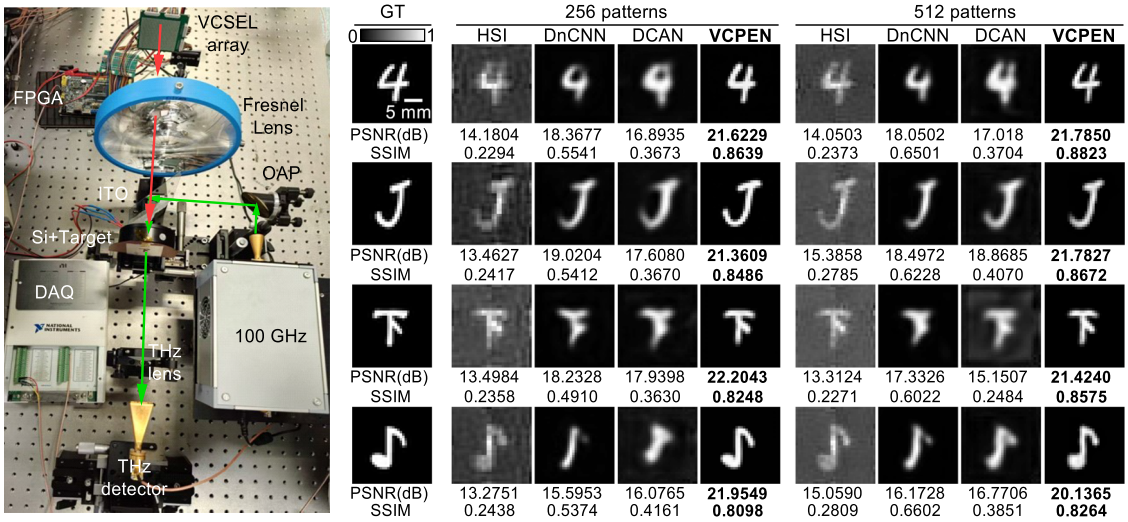


FIG. 5. Photograph of the experimental setup is shown on the left, with key components and optical path annotated. Experimental reconstruction of 32 × 32 pixel metal targets using different methods under β = 25% (256 patterns) and β = 50% (512 patterns). Quantitative metrics (PSNR/SSIM) are provided for each case, with the best performance highlighted in bold.

metal samples. In contrast, the VCPEN system faithfully preserves the sharp edges and fine structures of the characters while effectively eliminating background noise. To perform an objective quantitative evaluation of imaging quality, we used the original binary design patterns of the laser-engraved metal samples, processed via spatial registration and downsampling, as the ground truth. Based on the PSNR and SSIM calculated using this benchmark, the results further confirm that the VCPEN system significantly outperforms the comparison methods in both sampling conditions. Notably, even at the high sampling rate of 50 fps, VCPEN achieves a reconstruction PSNR exceeding 20 dB and an SSIM greater than 0.8 for all targets. This fully validates the system's ability to effectively overcome the dual challenges of hardware noise and low sampling rate in real-world THz imaging scenarios, achieving the optimal balance between detail restoration and noise robustness.

To further assess the VCPEN system's capability for real-time video imaging under dynamic conditions, we constructed an experimental platform for continuous acquisition of a rotating target. A precision motor was used to rotate a hollow metal letter "C" at controlled speeds between 2.5 and 5 rps. Video streams were captured using two configurations: 50 fps (256 patterns) and 25 fps (512 patterns). Representative frames extracted from the continuous imaging sequences at different rotation speeds and frame rates are shown in Fig. 6 (Multimedia available online), with particular emphasis on video segments recorded at 5 rps.

The results highlight the critical role of temporal resolution in dynamic imaging. At low rotation speeds, both configurations produced clear reconstructions of the letter's profile. However, when the speed increased to 5 rps, the 25 fps mode (single-frame acquisition time of 40 ms) exhibited pronounced motion blur and ghosting artifacts, as the sampling interval exceeded the characteristic time scale of target motion. In contrast, the 50 fps mode (single-frame acquisition time of 20 ms), despite its lower sampling ratio, effectively "froze" the motion due to its higher temporal resolution and accurately preserved the letter's spatial structure and edges. Continuous video playback under this configuration appears smooth and stable.

These observations demonstrate the VCPEN system's strong adaptability in high-speed dynamic scenarios. Operating at 50 fps, the system achieves an effective balance between reconstruction quality and temporal responsiveness, offering practical support for real-time THz video monitoring and dynamic process analysis. In practical systems, non-ideal effects of the VCSEL array, such as emission non-uniformity and temporal fluctuations, may affect pattern fidelity. In this study, these effects are mitigated by differential acquisition, DGI normalization, and learning-based reconstruction. Since all methods are evaluated under the same conditions, such effects do not affect the relative performance comparison. Explicit calibration will be considered in future work.

C. Discussion

Worthy of further discussion is that the photoresponse characteristic of the THz modulator is the key physical factor determining the upper limit of the VCPEN system's imaging rate. In our system, the high-resistivity silicon wafer acts as the spatial modulator, and its absorption modulation of the THz wave follows the Drude model: the density of photoexcited free carriers determines the plasma frequency, which in turn changes the material's complex permittivity and THz transmittance. To achieve high-contrast modulation, the illumination time must be long enough for the carrier generation and recombination processes to reach a steady state, thereby establishing a sufficient modulation depth. To systematically evaluate the temporal constraint of carrier lifetime on imaging quality, this study selected three high-resistivity silicon wafers with different carrier lifetimes, measured by MDPspot equipment to be 12, 26, and 100 μs.

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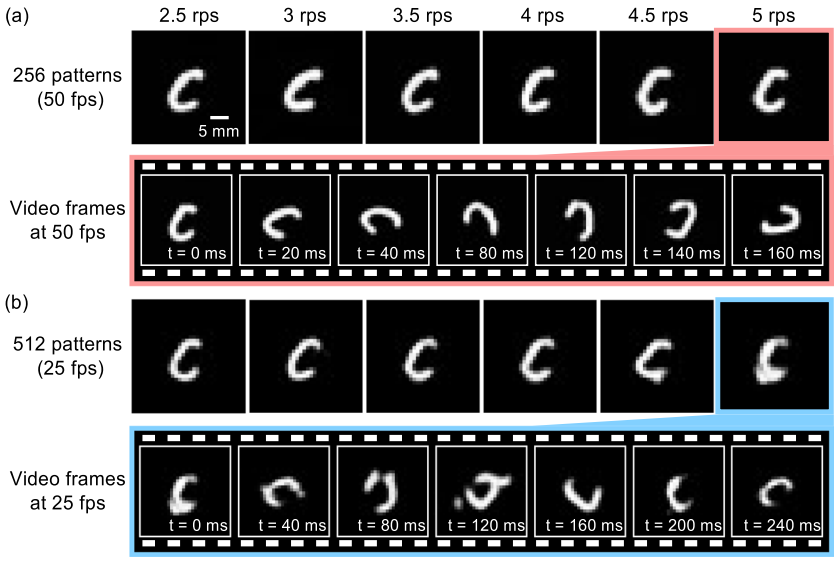


FIG. 6. Dynamic imaging performance comparison under varying frame rates. Representative frames extracted from continuous imaging sequences of a laser-engraved metal target (letter "C") rotating at speeds from 2.5 to 5 rps are shown, all at a resolution of 32 × 32 pixels. (a) Results from the 50 fps configuration (256 patterns), along with the consecutive video frames captured at a rotation speed of 5 rps. (b) Results from the 25 fps configuration (512 patterns), accompanied by the corresponding video sequence at 5 rps (Multimedia available online).

The three wafers were individually tested in the system using two sampling configurations (256 patterns and 512 patterns) at different single projection times (t_0), covering the full dynamic range from under-modulation to saturation.

As shown in Figs. 7(a)–7(c), we observed a distinct critical threshold behavior in imaging quality as the projection time varies. When the projection time is shorter than the carrier lifetime ($t_0 < T_{life}$), photo-generated carriers are in a transient accumulation stage, failing to reach the steady-state concentration peak required to effectively shield the THz wave. This is reflected in the reconstructed images by not only reduced SNR but also severe loss of structural information (side regions inside the red box in the figure), where

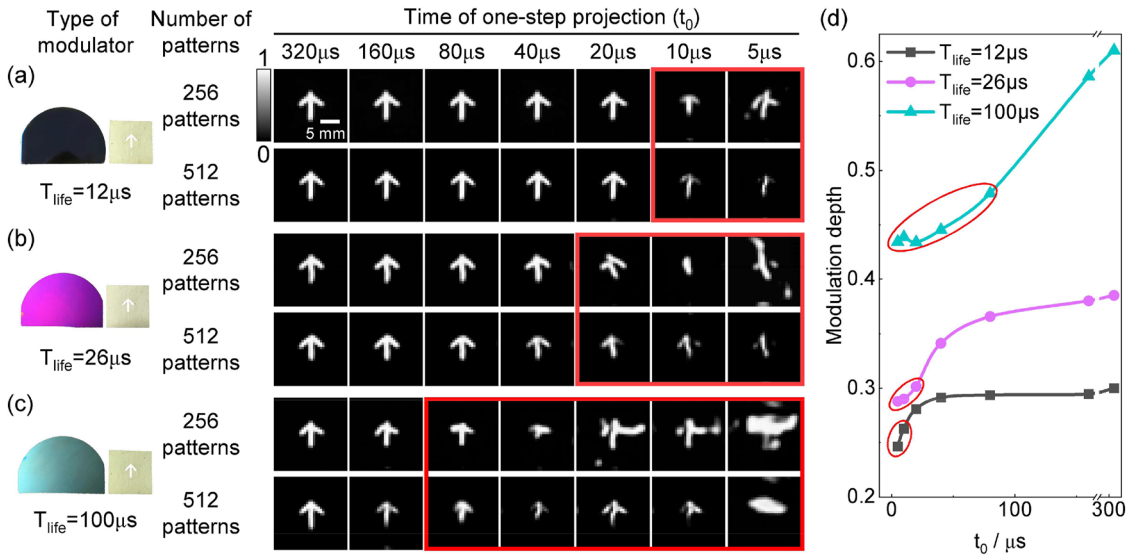


FIG. 7. Impact of modulator carrier lifetime on VCPEN imaging quality and temporal resolution. To evaluate the physical limits of the VCPEN system's modulation speed, three high-resistivity silicon wafer modulators with different carrier lifetimes (T_{life}) were tested: (a) 12 μs, (b) 26 μs, and (c) 100 μs. The system operated under two sampling configurations (256 and 512 patterns) with a fixed target position, testing single-step projection times (t_0) ranging from 320 μs down to 5 μs. The red boxes highlight conditions where the projection time is shorter than the carrier lifetime ($t_0 < T_{life}$), while regions outside the red boxes represent conditions where $t_0 > T_{life}$. (d) Modulation depth as a function of the projection time t_0 for three silicon wafers with different carrier lifetimes. A broken x axis is used to emphasize the rapid modulation-depth variation in the short projection-time regime.

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TABLE II. Comparison of frame rate and image performance of different modulation methods in THz SPI.

Methods	Pattern	Resolution	Frame rate (fps)	SSIM	PSNR (dB)	Reference
ATR + DMD	Hadamard	32 × 32	6	0.668	···	2020[26]
Metamaterial	Hadamard	8 × 8	5.5	0.72	30	2022[47]
STE + DMD	FMH	64 × 64	0.0195	···	22.70	2024[48]
MEH-PPV + DMD	Hadamard	16 × 16	25	1.00	···	2025[16]
ATR + DMD	Hadamard	64 × 64	7.58	0.60	···	2026[49]
Si + DMD	Hadamard	32 × 32	30	0.28	15.20	2025[11]
Si + VCSEL	Hadamard	32 × 32	25/50	0.26/0.24	14.45/13.60	2025[46]
Si + VCSEL	**Deep learning**	**32×32**	**25/50**	**0.86/0.84**	**21.28/21.78**	**This study**

the effective imaging information is overwhelmed by the transient noise of carrier evolution. Conversely, when the projection time exceeds the carrier lifetime ($t_0 > T_{life}$), the carrier distribution enters the steady-state saturation region, establishing sufficient modulation depth and contrast, allowing the reconstructed image to clearly restore the target profile (regions outside the red box in the figure). This result reveals the time-quality trade-off relationship of the VCPEN system, indicating that the modulator carrier lifetime is the core physical constraint limiting the system's maximum frame rate. To further quantify the effect of carrier lifetime on modulation performance, we measure the modulation depth as a function of the projection time t_0 for silicon wafers with different carrier lifetimes, as shown in Fig. 7(d). It can be observed that when t_0 is shorter than the carrier lifetime, the modulation depth is significantly reduced, indicating that the photocarrier density does not reach a steady state. As t_0 increases, the modulation depth gradually saturates, with longer carrier lifetimes corresponding to higher achievable modulation depth. Therefore, in the pursuit of next-generation ultrahigh-speed imaging, beyond identifying semiconductor materials with shorter carrier lifetimes (e.g., GaAs or ion-implanted silicon), achieving a high modulation depth remains a critical pathway to preventing modulation performance degradation.

To quantitatively illustrate the comprehensive performance advantages of our VCPEN system in THz single-pixel imaging, a detailed comparison of frame rate and key image quality metrics with other THz SPI imaging modulation methods is summarized in Table II.

The current system focuses on a 32 × 32 spatial resolution as a proof-of-concept for video-rate THz imaging. Scaling to higher resolutions (e.g., 64 × 64 or 128 × 128) is subject to several practical constraints. First, larger VCSEL arrays and higher-quality optical projection are required to maintain spatial fidelity and uniformity. Second, higher resolution generally demands more measurement patterns, which reduces the achievable frame rate under fixed projection time. Third, a smaller effective pixel size leads to reduced modulation contrast and a lower signal-to-noise ratio. In addition, the increased dimensionality also poses challenges for pattern optimization and reconstruction under compressed sampling conditions. These factors need to be jointly addressed in the future study to enable higher-resolution THz single-pixel imaging.

IV. CONCLUSION

In this paper, we proposed and experimentally validated VCPEN, a THz single-pixel video imaging framework based on the synergy of a VCSEL array and deep learning. Through joint hardware–software optimization, the system effectively reconciles the trade-off between real-time performance and imaging quality in conventional THz imaging. At the hardware level, the VCSEL array, leveraging its microsecond-scale ultra-high switching speed, removes the refresh-rate limitation associated with commonly used DMD-based optical pattern generation. Algorithmically, we designed a physics-enhanced deep learning network that generates parameter-efficient patterns highly adapted to the hardware's row-column driving mechanism via a Rank-1 matrix synthesis strategy. Simultaneously, it achieves high-fidelity image restoration at low sampling rates through a U-Net reconstruction network. The system achieved significant breakthroughs in dynamic imaging experiments, successfully realizing high-frame-rate video acquisition capabilities at 50 fps while maintaining superior imaging quality (PSNR >20 dB, SSIM >0.8) at this rate. Concurrently, this study revealed the physical constraint imposed by the carrier lifetime of the modulation material on the system's imaging rate, clarifying the direction for material optimization in the pursuit of future ultrafast imaging. The systemic methodology presented in this study not only provides a feasible hardware–software collaborative scheme for realizing real-time, high-quality THz SPI but also offers valuable insights for the development of high-speed SPI technologies in other spectral bands, such as the visible range.

SUPPLEMENTARY MATERIAL

See the supplementary material for details of the control signal generation and level logic (Sec. S1), neural network architecture (Sec. S2), differentiable optimization via the straight-through estimator (Sec. S3), and code availability (Sec. S4).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Junjie Huo: Data curation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jing Wang**: Data curation (equal); Investigation (equal). **Rongbin She**: Resources (equal); Software (equal); Supervision (equal); Validation (equal). **Yuanfu Lu**: Resources (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1]Z. Yan, L.-G. Zhu, K. Meng, W. Huang, and Q. Shi, "THz medical imaging: From *in vitro* to *in vivo*," Trends Biotechnol. **40**, 816–830 (2022).
[2]A. Ren, A. Zahid, D. Fan, X. Yang, M. A. Imran, A. Alomainy, and Q. H. Abbasi, "State-of-the-art in terahertz sensing for food and water security—A comprehensive review," Trends Food Sci. Technol. **85**, 241–251 (2019).
[3]Q. Sun, Y. He, K. Liu, S. Fan, E. P. J. Parrott, and E. Pickwell-MacPherson, "Recent advances in terahertz technology for biomedical applications," Quant. Imaging Med. Surg. **7**, 345–355 (2017).
[4]Y.-C. Shen, "Terahertz pulsed spectroscopy and imaging for pharmaceutical applications: A review," Int. J. Pharm. **417**, 48–60 (2011).
[5]H. Zhong, A. Redo-Sanchez, and X.-C. Zhang, "Identification and classification of chemicals using terahertz reflective spectroscopic focal-plane imaging system," Opt. Express **14**, 9130 (2006).
[6]X. Li, J. Li, Y. Li, A. Ozcan, and M. Jarrahi, "High-throughput terahertz imaging: Progress and challenges," Light: Sci. Appl. **12**, 233 (2023).
[7]G. M. Gibson, S. D. Johnson, and M. J. Padgett, "Single-pixel imaging 12 years on: A review," Opt. Express **28**, 28190 (2020).
[8]Y. Peng, C. Shi, Y. Zhu, M. Gu, and S. Zhuang, "Terahertz spectroscopy in biomedical field: A review on signal-to-noise ratio improvement," PhotoniX **1**, 12 (2020).
[9]Q. Hu, X. Wei, Y. Pang, and L. Lang, "Advances on terahertz single-pixel imaging," Front. Phys. **10**, 982640 (2022).
[10]X. Yang, Z. Tian, X. Chen, M. Hu, Z. Yi, C. Ouyang, J. Gu, J. Han, and W. Zhang, "Terahertz single-pixel near-field imaging based on active tunable subwavelength metallic grating," Appl. Phys. Lett. **116**, 241106 (2020).
[11]R. She, X. Luo, J. Huo, D. Wei, and Y. Lu, "Terahertz subwavelength single-pixel video based on computational imaging," Opt. Express **33**, 45281 (2025).
[12]W. L. Chan, K. Charan, D. Takhar, K. F. Kelly, R. G. Baraniuk, and D. M. Mittleman, "A single-pixel terahertz imaging system based on compressed sensing," Appl. Phys. Lett. **93**, 121105 (2008).
[13]H. Shen, L. Gan, N. Newman, Y. Dong, C. Li, Y. Huang, and Y. C. Shen, "Spinning disk for compressive imaging," Opt. Lett. **37**, 46 (2012).
[14]I. R. Hooper, N. E. Grant, L. E. Barr, S. M. Hornett, J. D. Murphy, and E. Hendry, "High efficiency photomodulators for millimeter wave and THz radiation," Sci. Rep. **9**, 18304 (2019).
[15]R. I. Stantchev, B. Sun, S. M. Hornett, P. A. Hobson, G. M. Gibson, M. J. Padgett, and E. Hendry, "Noninvasive, near-field terahertz imaging of hidden objects using a single-pixel detector," Sci. Adv. **2**, e1600190 (2016).
[16]J. Liang, J. Zhang, Z. Wang, R. Wang, Z. Yao, R. Singh, Z. Tian, and W. Zhang, "Photoactive polymer-silicon heterostructures for terahertz spatial light modulation and video-rate single-pixel compressive imaging," Adv. Funct. Mater. **35**, 2422478 (2025).
[17]L. Wang, Y. Zhang, X. Guo, T. Chen, H. Liang, X. Hao, X. Hou, W. Kou, Y. Zhao, T. Zhou, S. Liang, and Z. Yang, "A review of THz modulators with dynamic tunable metasurfaces," Nanomaterials **9**, 965 (2019).
[18]Y. Bai, T. Bu, K. Chen, and S. Zhuang, "Review about the optical-controlled terahertz waves modulator," Appl. Spectrosc. Rev. **50**, 707–727 (2015).
[19]J. Liang, J. Zhang, and Z. Tian, "Video-rate terahertz single-pixel super-resolution imaging," in *2024 49th International Conference on Infrared, Millimeter, and Terahertz Waves (IRMMW-THz)* (IEEE, Perth, 2024), pp. 1–2.
[20]A. Kannegulla, M. I. B. Shams, L. Liu, and L.-J. Cheng, "Photo-induced spatial modulation of THz waves: Opportunities and limitations," Opt. Express **23**, 32098 (2015).
[21]L. Zanotto, R. Piccoli, J. Dong, D. Caraffini, R. Morandotti, and L. Razzari, "Time-domain terahertz single-pixel imaging," SPIE **11685**, 116851D (2021).
[22]M.-J. Sun, M. P. Edgar, G. M. Gibson, B. Sun, N. Radwell, R. Lamb, and M. J. Padgett, "Single-pixel three-dimensional imaging with time-based depth resolution," Nat. Commun. **7**, 12010 (2016).
[23]S. Jiang, X. Li, Z. Zhang, W. Jiang, Y. Wang, G. He, Y. Wang, and B. Sun, "Scan efficiency of structured illumination in iterative single pixel imaging," Opt. Express **27**, 22499 (2019).
[24]Z. Zhang, X. Wang, G. Zheng, and J. Zhong, "Hadamard single-pixel imaging versus Fourier single-pixel imaging," Opt. Express **25**, 19619 (2017).
[25]R. She, W. Liu, Y. Lu, Z. Zhou, and G. Li, "Fourier single-pixel imaging in the terahertz regime," Appl. Phys. Lett. **115**, 021101 (2019).
[26]R. I. Stantchev, X. Yu, T. Blu, and E. Pickwell-MacPherson, "Real-time terahertz imaging with a single-pixel detector," Nat. Commun. **11**, 2535 (2020).
[27]W.-K. Yu and Y.-M. Liu, "Single-pixel imaging with origami pattern construction," Sensors **19**, 5135 (2019).
[28]D. L. Donoho, "Compressed sensing," IEEE Trans. Inf. Theory **52**, 1289–1306 (2006).
[29]L. Bian, J. Suo, Q. Dai, and F. Chen, "Experimental comparison of single-pixel imaging algorithms," J. Opt. Soc. Am. A **35**, 78 (2018).
[30]R. I. Stantchev, D. B. Phillips, P. Hobson, S. M. Hornett, M. J. Padgett, and E. Hendry, "Compressed sensing with near-field THz radiation," Optica **4**, 989 (2017).
[31]Z. Li, J. Huang, D. Shi, Y. Chen, K. Yuan, S. Hu, and Y. Wang, "Single-pixel imaging with untrained convolutional autoencoder network," Opt. Laser Technol. **167**, 109710 (2023).
[32]S. Lin, Y. Zhang, H. Wang, Z. Bo, F. Wang, and G. Situ, "Learning-based high-speed single-pixel imaging using a cyclic random mask," Opt. Express **33**, 25728 (2025).
[33]K. Song, Y. Bian, D. Wang, R. Li, K. Wu, H. Liu, C. Qin, J. Hu, and L. Xiao, "Advances and challenges of single-pixel imaging based on deep learning," Laser Photonics Rev. **19**, 2401397 (2025).
[34]M. Lyu, W. Wang, H. Wang, H. Wang, G. Li, N. Chen, and G. Situ, "Deep-learning-based ghost imaging," Sci. Rep. **7**, 17865 (2017).
[35]C. F. Higham, R. Murray-Smith, M. J. Padgett, and M. P. Edgar, "Deep learning for real-time single-pixel video," Sci. Rep. **8**, 2369 (2018).
[36]Y. Deng, R. She, W. Liu, Y. Lu, and G. Li, "High-efficiency terahertz single-pixel imaging based on a physics-enhanced network," Opt. Express **31**, 10273 (2023).
[37]Y.-L. Zhu, R.-B. She, W.-Q. Liu, Y.-F. Lu, and G.-Y. Li, "Deep learning optimized terahertz single-pixel imaging," IEEE Trans. Terahertz Sci. Technol. **12**, 165–172 (2022).
[38]K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," IEEE Trans. Image Process. **26**, 3142–3155 (2017).
[39]S. Li, C. H. Liu, Q. Lin, B. Xie, Z. Ding, G. Huang, and J. Tang, "Domain conditioned adaptation network," Proc. AAAI Conf. Artif. Intell. **34**(07), 11386–11393 (2020).

18 July 2026 12:32:56

© Author(s) 2026

[40]Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image restoration," IEEE Trans. Pattern Anal. Mach. Intell. **43**, 2480–2495 (2021).
[41]F. Wang, C. Wang, M. Chen, W. Gong, Y. Zhang, S. Han, and G. Situ, "Far-field super-resolution ghost imaging with a deep neural network constraint," Light: Sci. Appl. **11**, 1 (2022).
[42]Y. Zhu, S. Liu, X. Wang, R. Li, H. Yang, J. Wang, H. Zhu, Y. Ke, N. Xu, H. Chen, and S. Deng, "Deep learning empowered sub-diffraction terahertz backpropagation single-pixel imaging," ACS Photonics **12**, 5453–5463 (2025).
[43]Y. Zhu, S. Liu, X. Wang, R. Li, H. Yang, J. Wang, H. Zhu, Y. Ke, N. Xu, H. Chen, and S. Deng, "Deep learning empowered sub-diffraction terahertz backpropagation single-pixel imaging," ACS Photonics **12**(10), 5453–5463 (2025).
[44]Z.-H. Xu, W. Chen, J. Penuelas, M. Padgett, and M.-J. Sun, "1000 fps computational ghost imaging using LED-based structured illumination," Opt. Express **26**, 2427 (2018).
[45]H. Huang, L. Li, Y. Ma, and M. Sun, "25000 fps computational ghost imaging with ultrafast structured illumination," Electron. Mater. **3**, 93–100 (2022).
[46]R. She, J. Huo, D. Wei, and Y. Lu, "Terahertz single-pixel video based on a matrix-addressable infrared vertical-cavity surface-emitting laser array," Appl. Phys. Lett. **127**, 251101 (2025).
[47]W. Li, X. Hu, J. Wu, K. Fan, B. Chen, C. Zhang, W. Hu, X. Cao, B. Jin, Y. Lu, J. Chen, and P. Wu, "Dual-color terahertz spatial light modulator for single-pixel imaging," Light: Sci. Appl. **11**, 191 (2022).
[48]X. Chen, J. Li, L. Du, C. Sun, Z. Zhai, S. Zhong, Q. Liu, K. Huang, and L.-G. Zhu, "Near-field terahertz single-pixel imaging with ultralow sampling ratio," Opt. Express **32**, 35126 (2024).
[49]S. Mou, R. I. Stantchev, S. Saxena, H. Ou, S. Rane, S. L. Pain, J. D. Murphy, E. Hendry, J. Lloyd-Hughes, and E. Pickwell-MacPherson, "All-fibre-coupled terahertz single-pixel imaging for biomedical applications," Nat. Commun. **17**, 1571 (2026).

18 July 2026 12:32:56

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